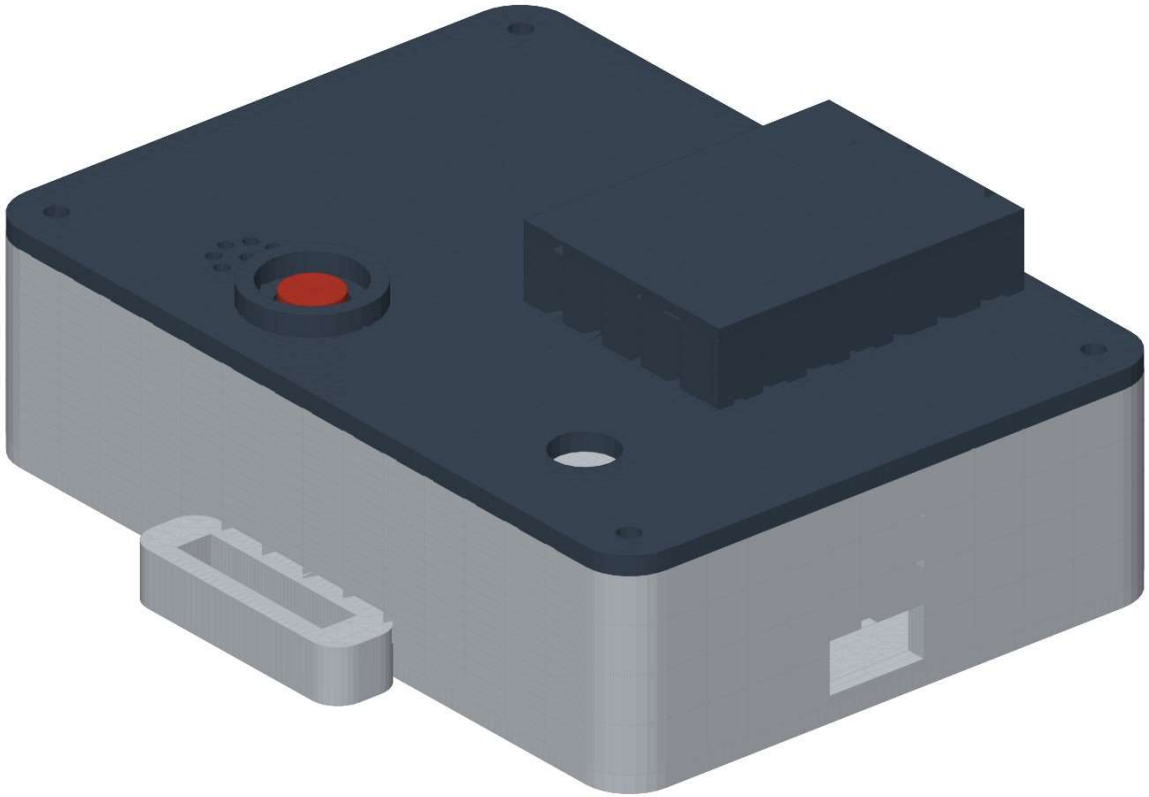


Overview

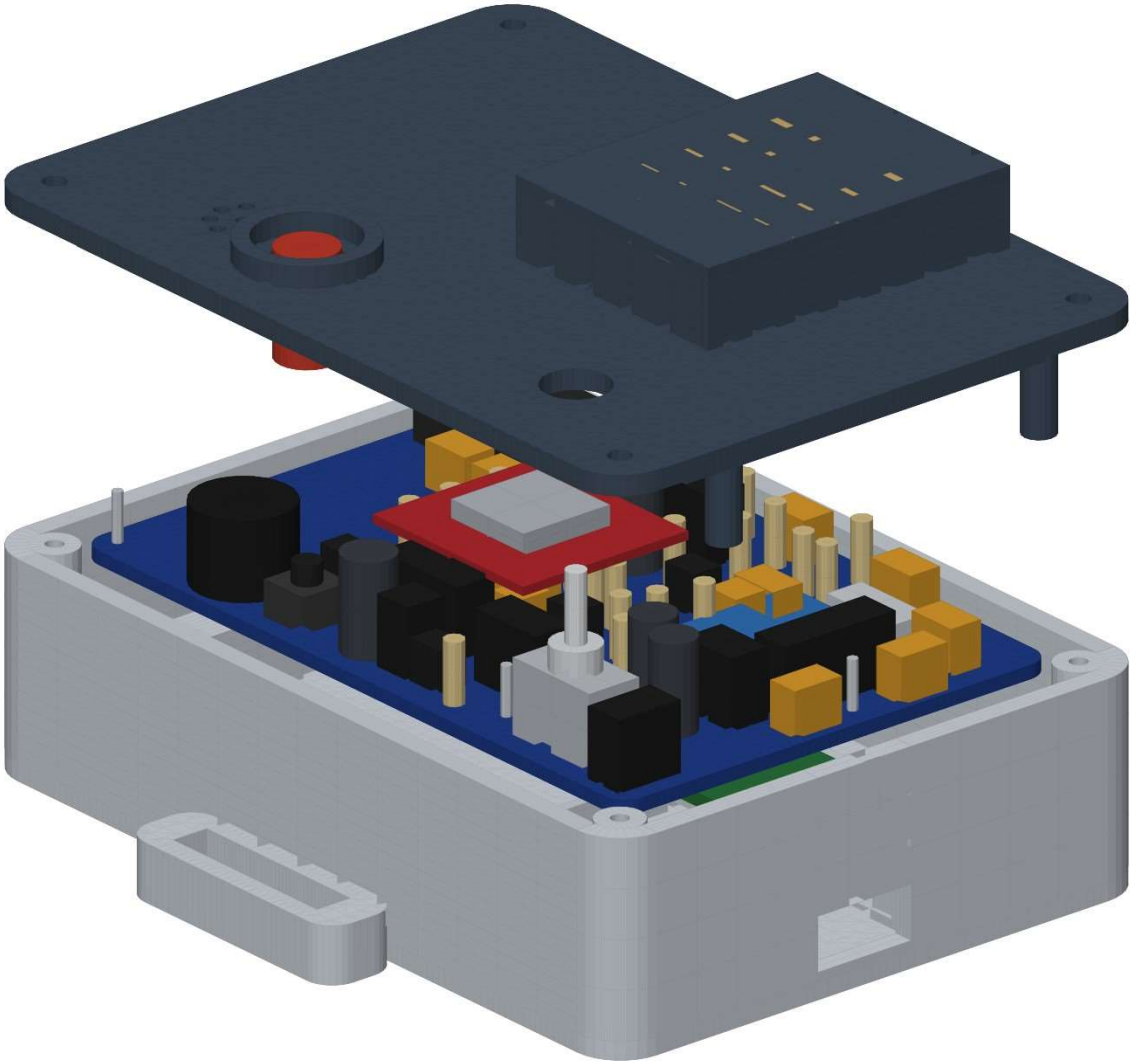
Three-part case for the 90 x 60 mm SAFE_LINK carrier PCB: base, lid and a printed SOS plunger. Two-piece shell closed with 4 x M2.5 screws.

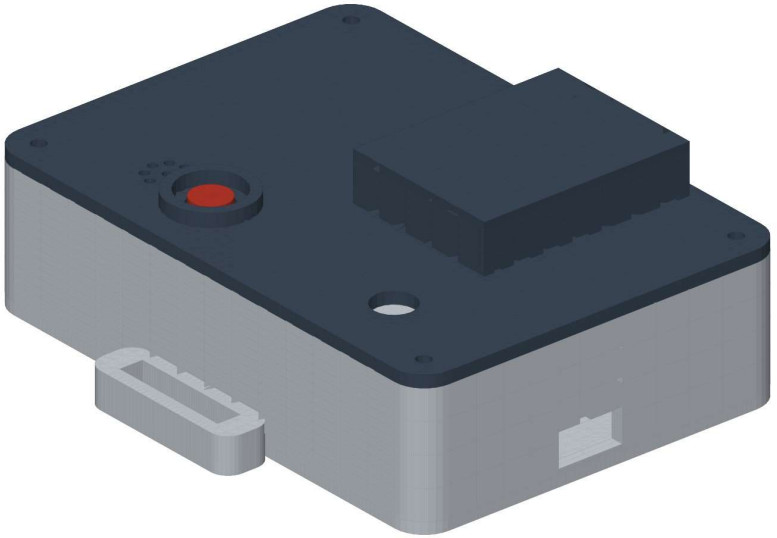
Item	Size (mm)
Outer footprint	x 75
Outer height (case)	28.4
Height with GPS dome	38.0
Inner cavity	x 70
PCB	x 60 x 1.6
Free height above PCB	13.7
Wall / floor / lid	2.5 / 2.2 / 2.4
Battery pocket	x 34 x 5.5 LiPo

Versions

A Armband: strap lugs front and back (slot 24 x 4.8) B Backpack / belt: two strap bars under the floor + lanyard tab
C Demo bench: large lid window, 4 rubber feet

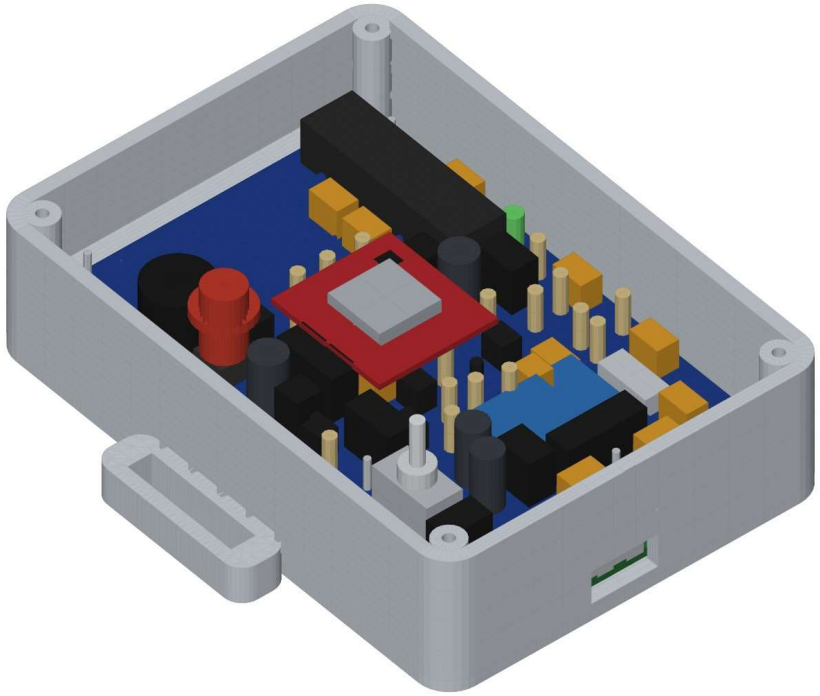
Board data from the EasyEDA placement (2026-10-18). Component heights are nominal parts; the case has 13.7 mm free height above the PCB for 11.5 mm capacitors and the SIM800L module.
Coordinates on the drawings: origin = front-left inner corner of the cavity, X to the right, Y toward the back.

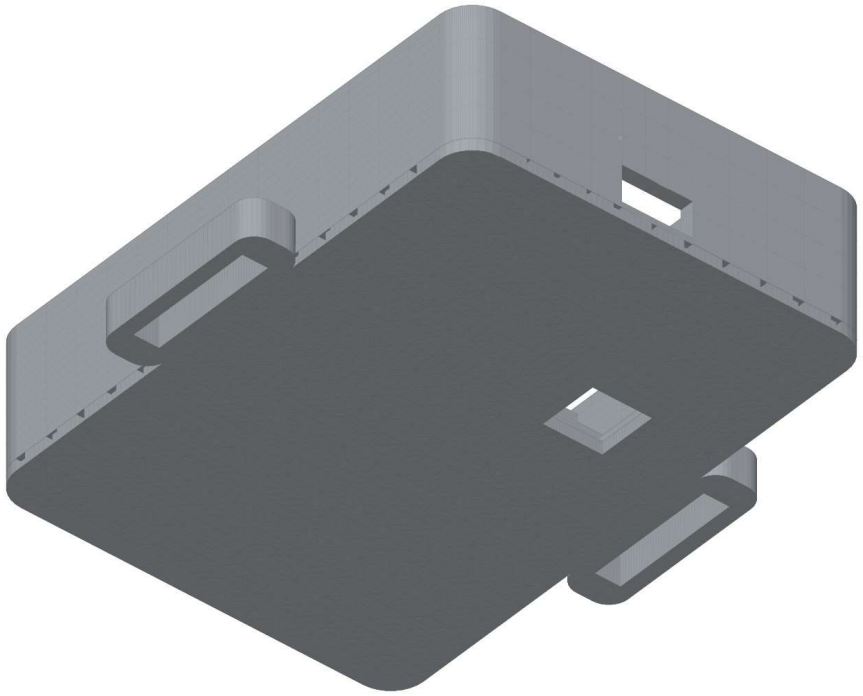


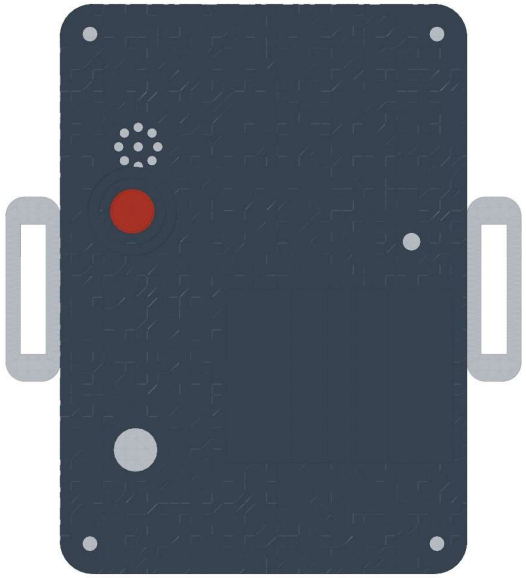


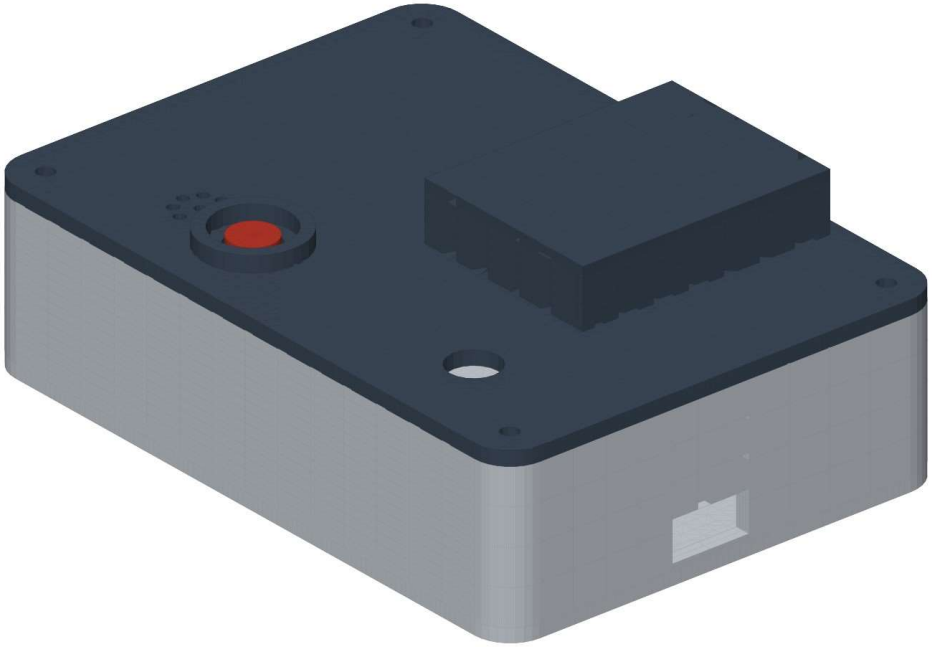
Features

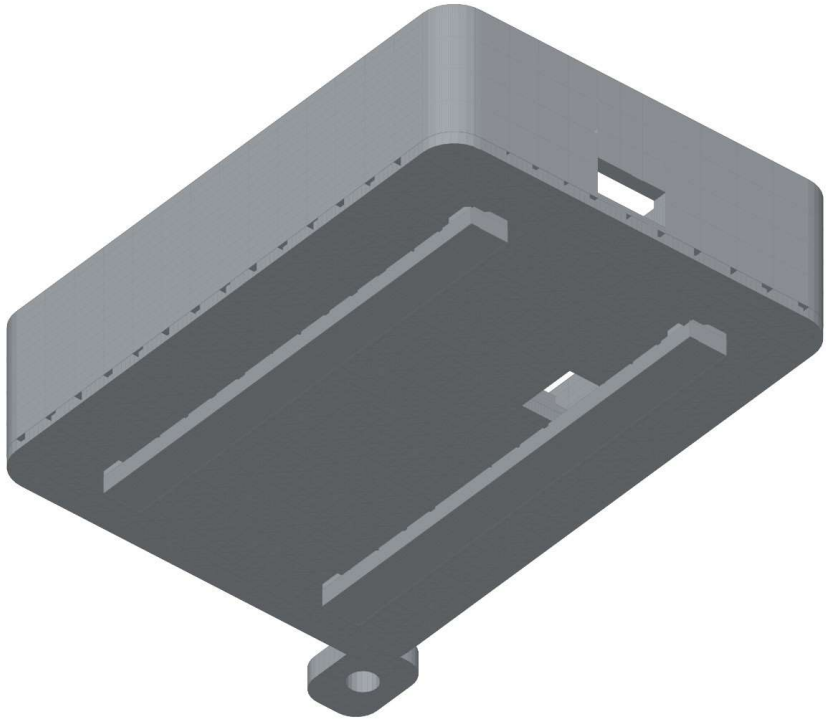
- Lugs: 34 x 10 x 7 mm, strap slot 24 x 4.8 mm
- USB-C opening on the right wall: 13 x 6.6 mm
- Skin-side heart-rate window 11 x 11 mm in the floor
- Lid: SOS plunger with raised guard ring, toggle-switch hole, LED hole, buzzer holes, 4 screw holes
- GPS module slides into the dome from below and rests on two ledges; patch faces the 1.6 mm roof
- 4 lid pressure pads press the PCB down onto the ledges

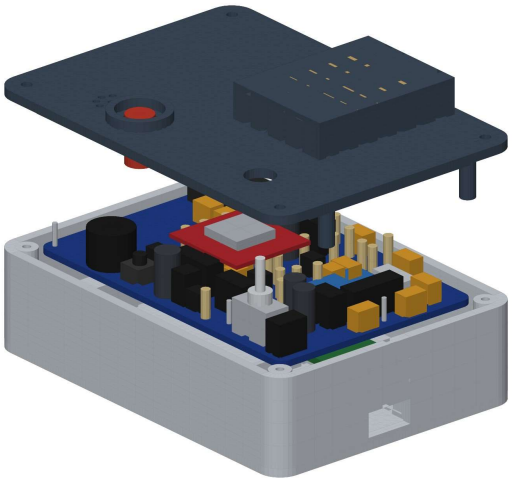






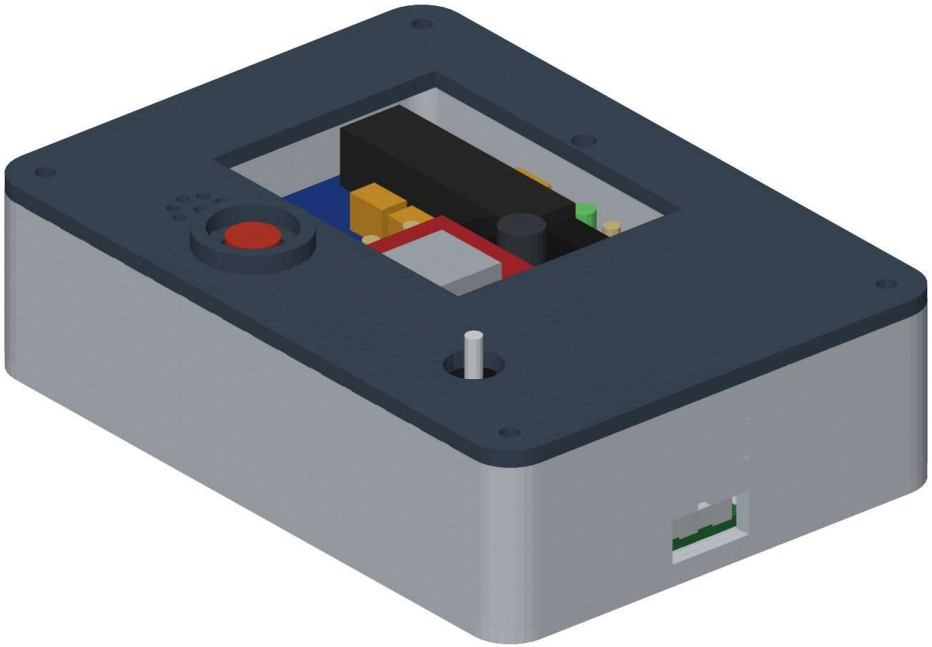


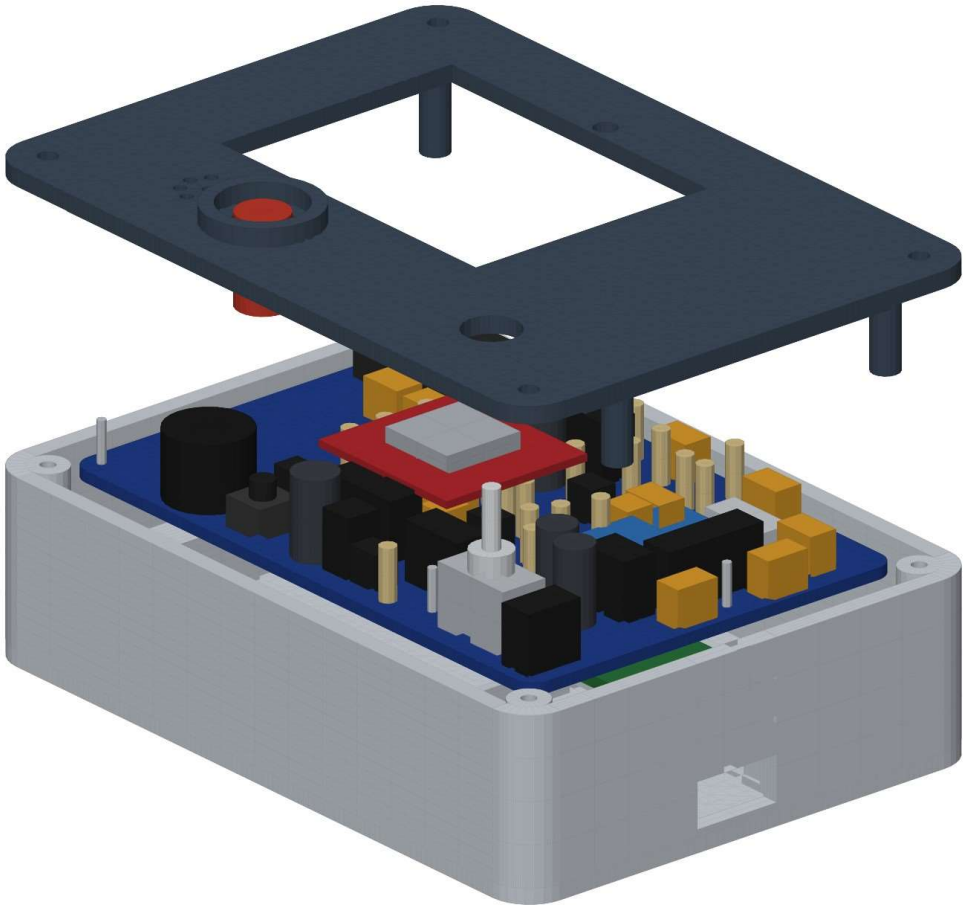




Features

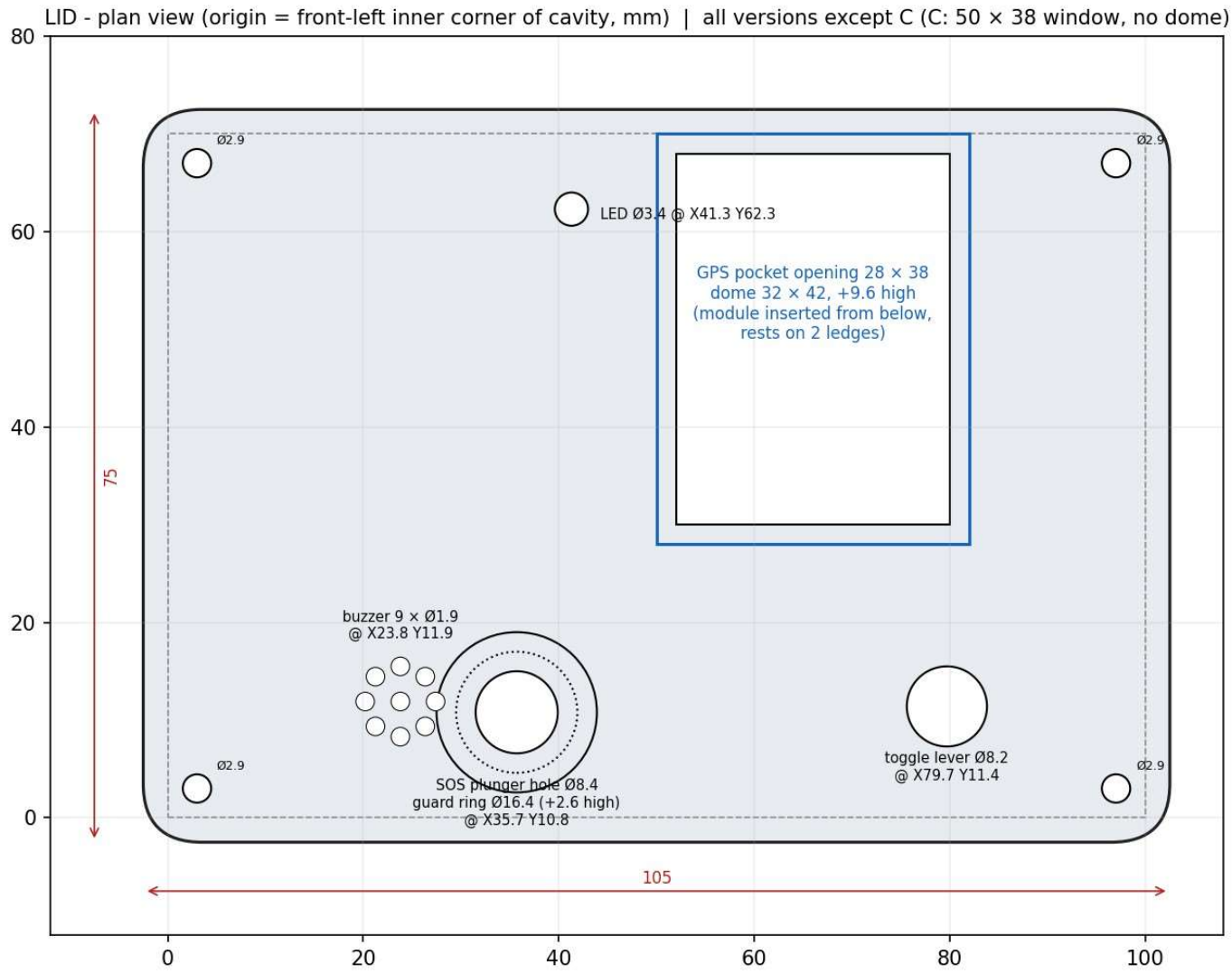
- Two bars 84 x 7 x 3 mm on legs, 3 mm gap for a webbing strap up to 26 mm wide
- Lanyard tab at the back-left corner, hole Ø5.2 mm
- Floor window for the heart-rate sensor stays in the middle of the base
- Everything else (lid, openings, USB-C, dome, SOS) is identical to version A

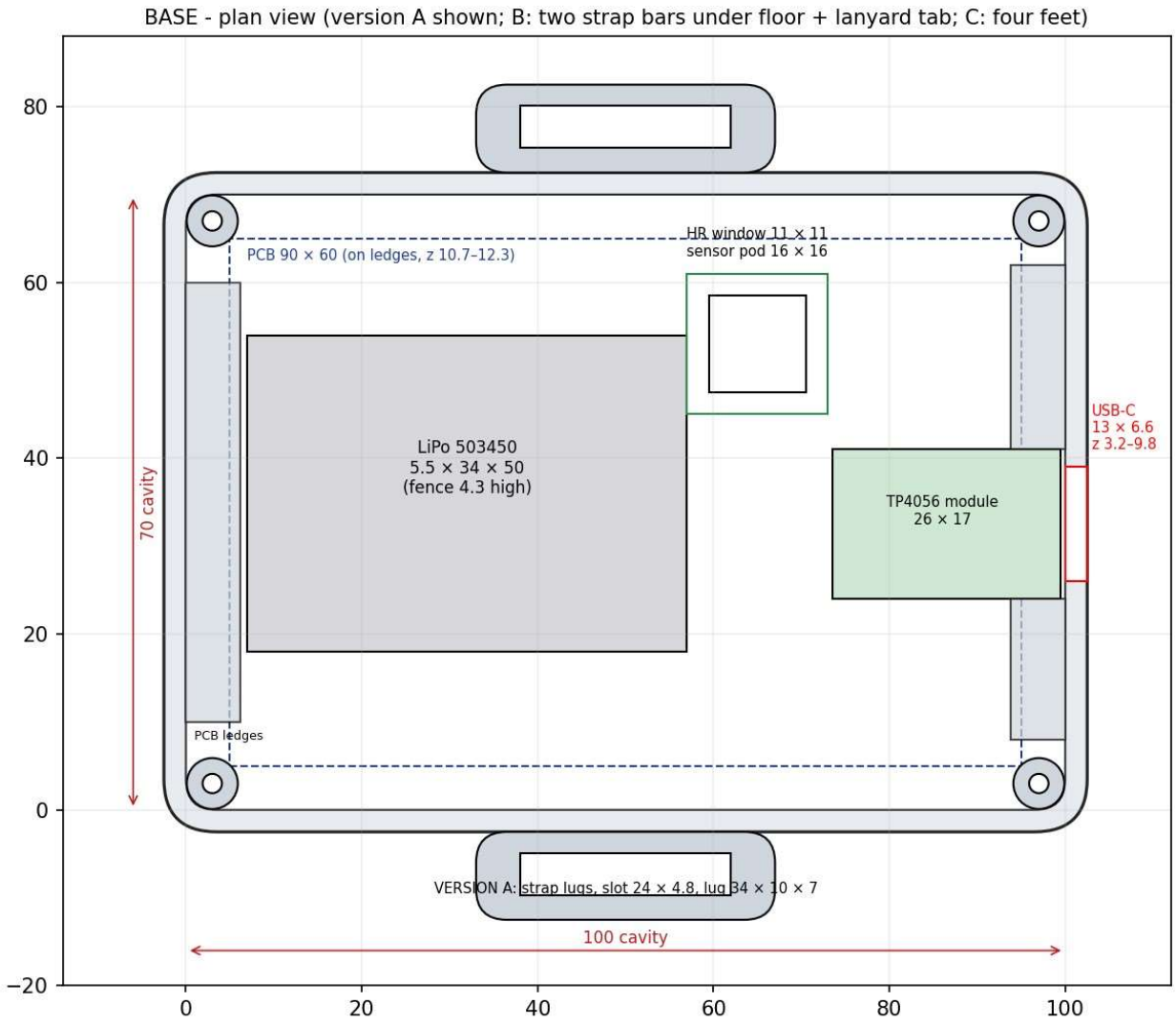




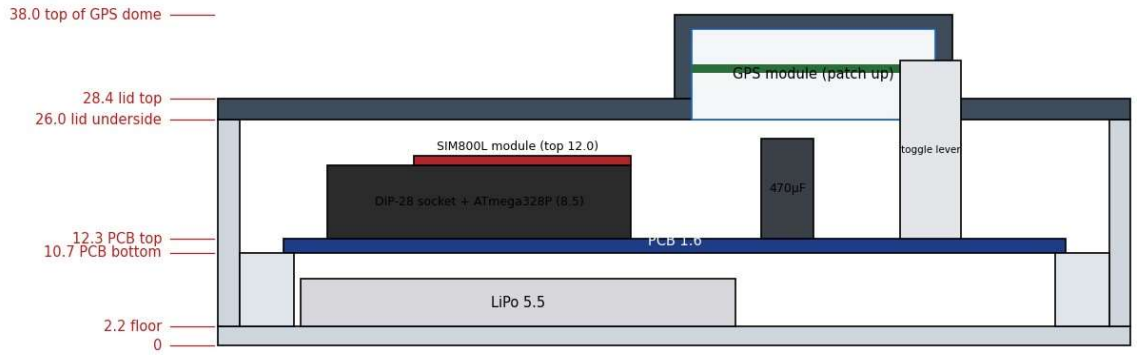
Features

- Lid window 50 x 38 mm shows the PCB; no GPS dome (GPS module placed on the board)
- Four Ø8 x 1.6 mm rubber feet on the underside
- Same base cavity, USB-C opening and SOS plunger as A and B


LID - plan view (origin = front-left inner corner of cavity, mm) | all versions except C (C: 50 × 38 window, no dome)
Ø2.9
Ø2.9
LED Ø3.4 @ X41.3 Y62.3
GPS pocket opening 28 × 38
dome 32 × 42, +9.6 high
(module inserted from below,
rests on 2 ledges)
buzzer 9 × Ø1.9
@ X23.8 Y11.9
Ø2.9
Ø2.9
toggle lever Ø8.2
@ X79.7 Y11.4
SOS plunger hole Ø8.4
guard ring Ø16.4 (+2.6 high)
@ X35.7 Y10.8
75
105
0
20
40
60
80
0
20
40
60
80
100


BASE - plan view (version A shown; B: two strap bars under floor + lanyard tab; C: four feet)
PCB 90 × 60 (on ledges, z 10.7–12.3)
HR window 11 × 11
sensor pod 16 × 16
LiPo 503450
5.5 × 34 × 50
(fence 4.3 high)
TP4056 module
26 × 17
USB-C
13 × 6.6
z 3.2–9.8
70 cavity
PCB ledges
VERSION A: strap lugs, slot 24 × 4.8, lug 34 × 10 × 7
100 cavity
80
60
40
20
0
−20
0
20
40
60
80
100

SECTION through the middle of the case (X-Z, mm). Free height above PCB 13.7; walls 2.5; floor 2.2; lid 2.4



Level	Z (mm)	Level	Z (mm)
Floor top	2.2	Lid underside	26.0
Battery top	7.7	Lid top	28.4
PCB bottom	10.7	Toggle lever tip	32.8
PCB top	12.3	GPS dome roof	38.0

Part	Qty	Notes
Base (A, B or C)	1	PETG or ASA
Lid (A/B or C)	1	PETG or ASA, same material as base
SOS plunger	1	print upright, red PETG
M2.5 x 12 self-tapping screw	4	lid to the 4 corner bosses (pilot Ø2.2)
LiPo 3.7 V, 1000 mAh, 503450 size	1	protected cell, 5.5 x 34 x 50 mm
TP4056 module (USB-C, protected)	1	26 x 17 mm, sits on 2 rails, USB-C at the right wall
Heart-rate pod (MAX30102 board)	1	16 x 16 mm, face down on the window frame
GPS module (NEO-6M type)	1	25 x 35 mm, patch up, in the dome (A, B)
Strap 24 mm / webbing 26 mm	1	A: through both lugs; B: under both bars
Rubber feet Ø8 x 1.6	4	version C only

Print settings

- Material PETG or ASA (PLA softens in a hot car); 0.2 mm layers, 3 walls, 25% infill
- Base: open side up, no supports. Lid: top side up, supports under the dome roof (bridge 28 mm), or print the roof with 100% infill
- Do not use supports in the screw holes; drill Ø2.2 if they print tight
- Scale 100%; STL units are millimetres

Assembly order

1 Fit the TP4056 module on its rails, USB-C into the right wall opening.
2 Place the LiPo in its pocket and the heart-rate pod over the window.
3 Rest the PCB on the left/right ledges (components up).
4 Plug the SIM800L and GY-521 modules into their headers, run the HR and GPS cables.
5 Slide the GPS module into the dome and fit the SOS plunger.
6 Close the lid and fit the 4 screws. Toggle lever and plunger protrude through the lid.